Cloud-Aware HAP Deployment for Quantum Group Key Distribution over the Tohoku Region

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Outline

I. Introduction
II. Goal
III. System
IV. Simulation

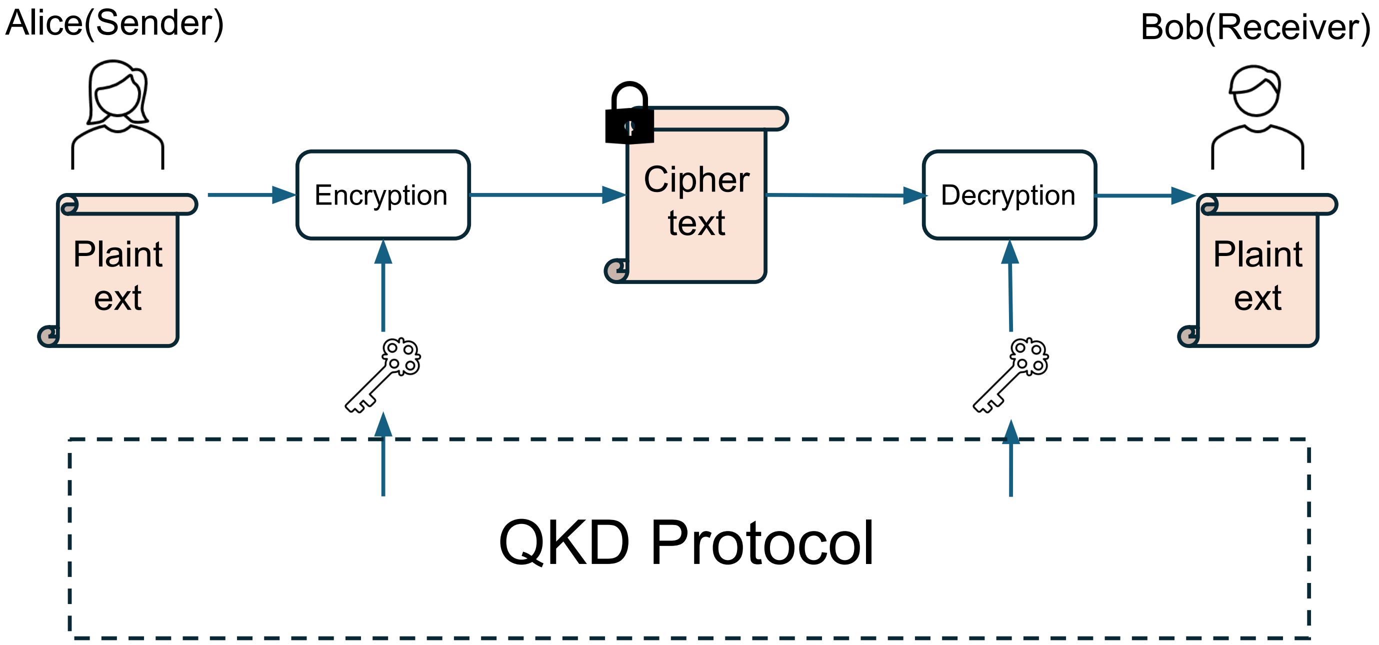
Background/motivation

- The Role of Local Governments During Disasters
  - Government agencies play a crucial role as communication hubs for coordinating communication between different regions during a disaster.
  - They handle highly confidential operational data, such as real-time damage assessments and recovery instructions.

- The Threat of Quantum Computing
  - Traditional encryption methods (e.g., RSA) are threatened by advances in quantum computing.
  - Securing these critical disaster management hubs is an urgent issue that must be implemented proactively

Background/motivation

Quantum Key Distribution is a technology that is secure based on the laws of physics.



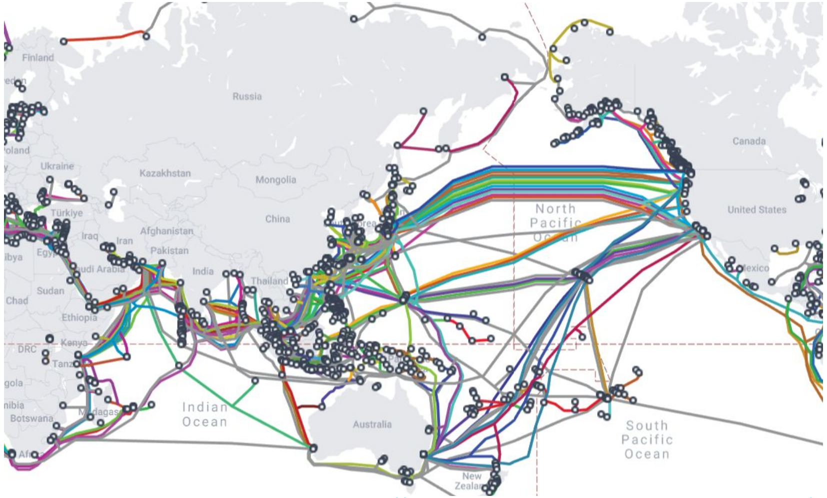
Background/motivation

Conventional: Optical fiber

- Vulnerable to ground-level damage
- physical disconnection during disasters

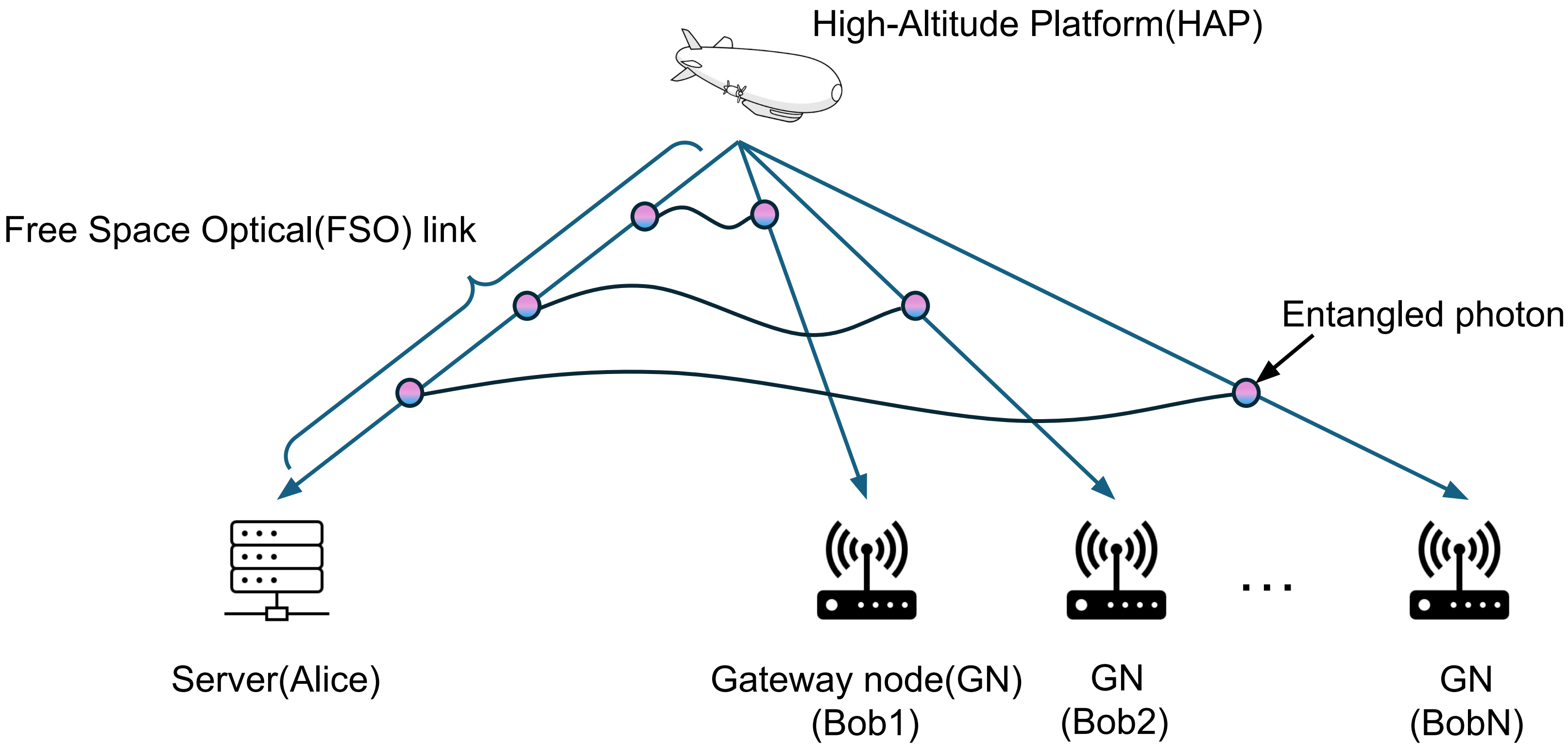
New: High Altitude Platforms

- Floating in the stratosphere (~20 km high)
- Unaffected by earthquakes



https://www.submarinecablemap.com/

Background/motivation



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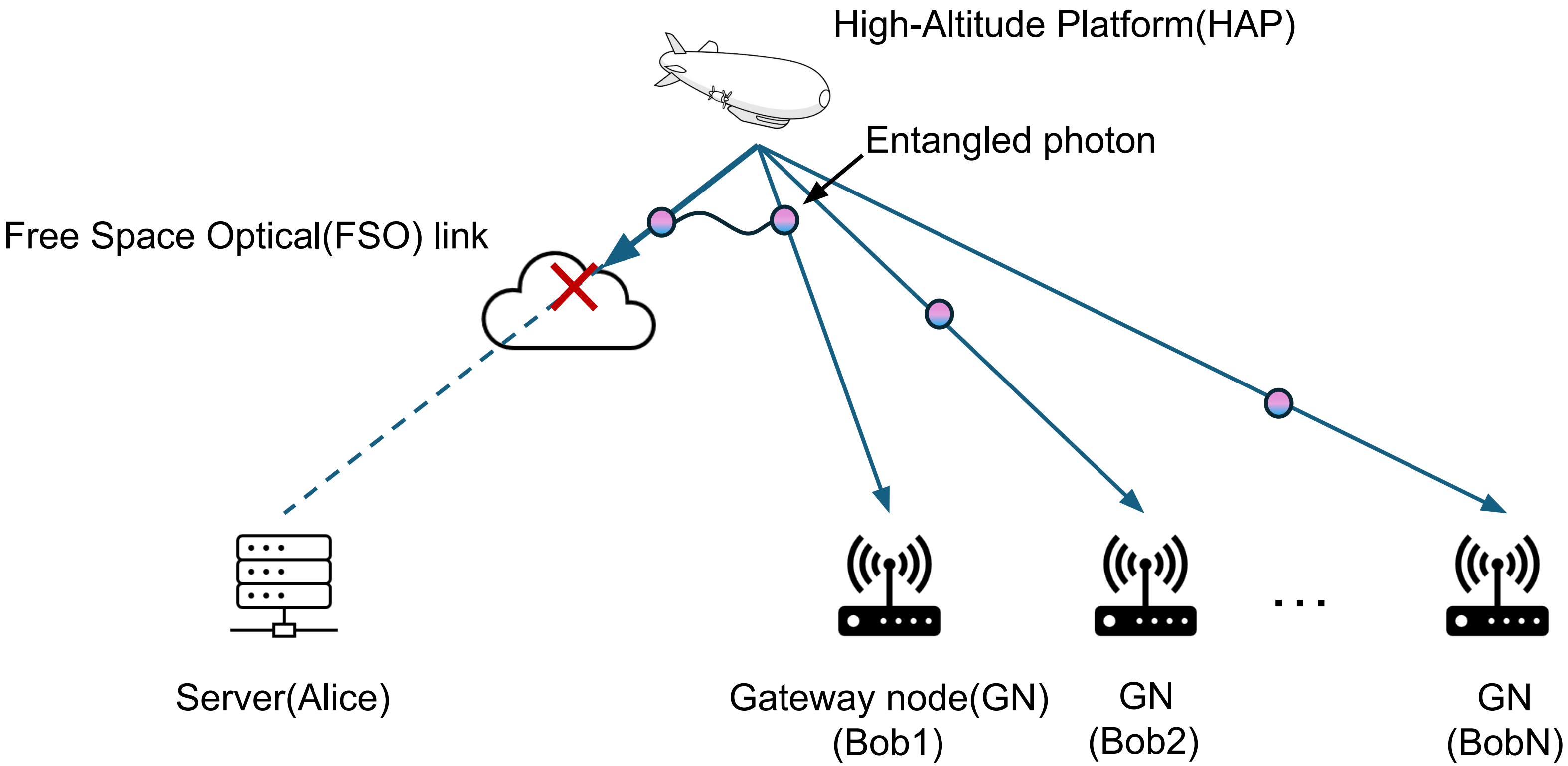

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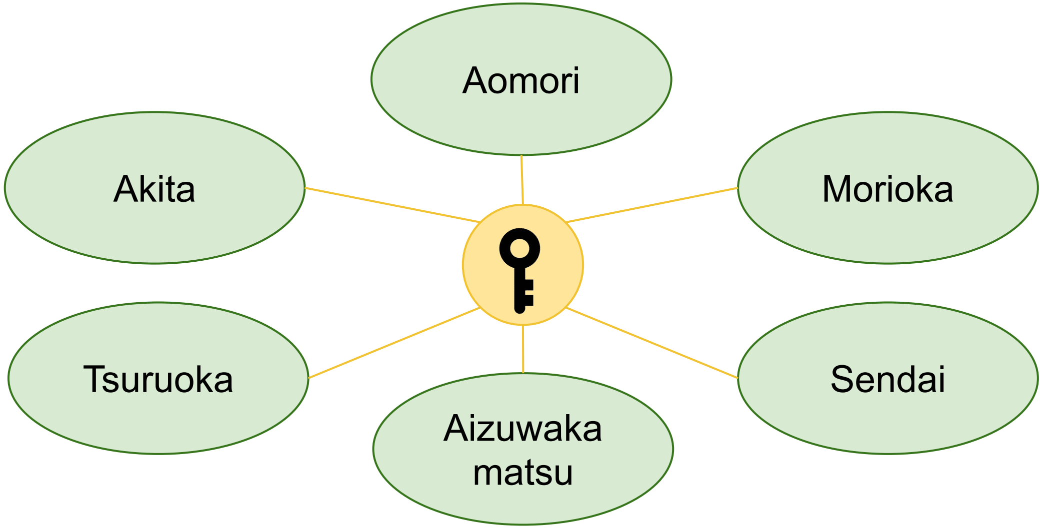
Goal

All 6 cities in Tohoku obtain the group key.

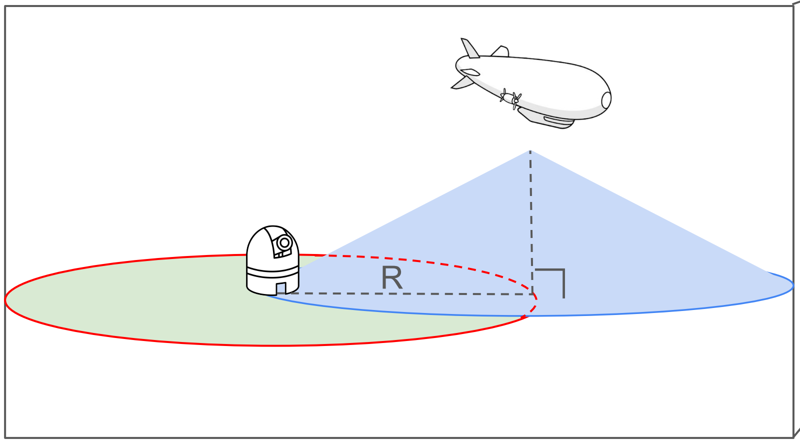
- To build a QKD network using HAP that takes cloud cover into account in the Tohoku region.

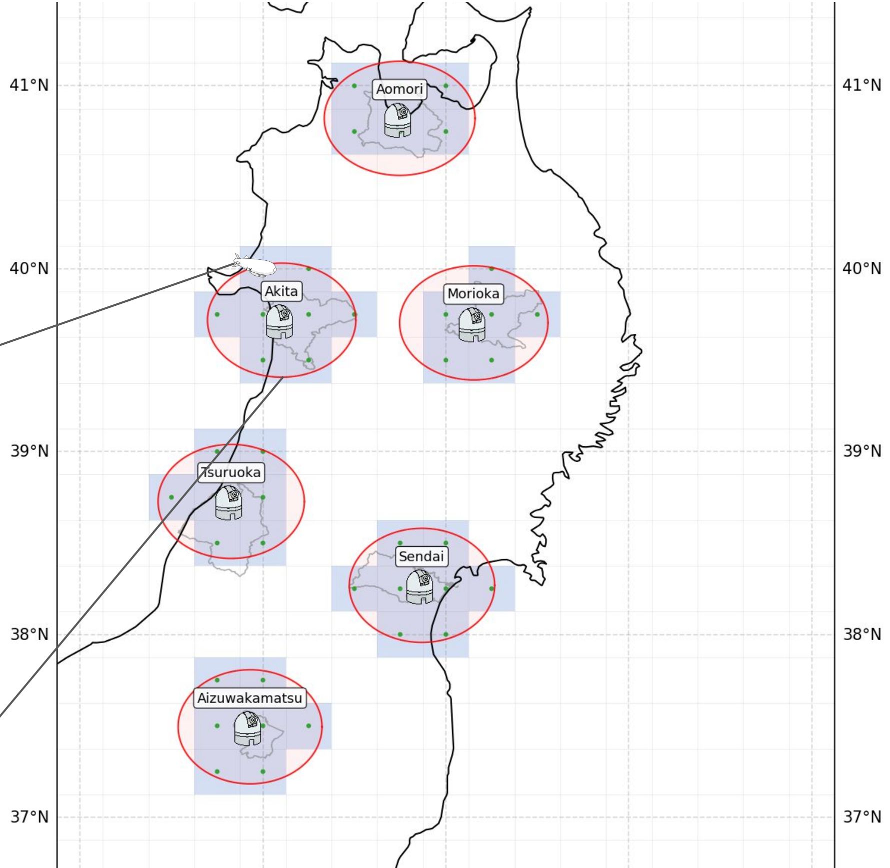


Outline

Defining HAP candidate positions

zenith angle = 60°
- R = 34.6 km
- 6 ~ 8 grids

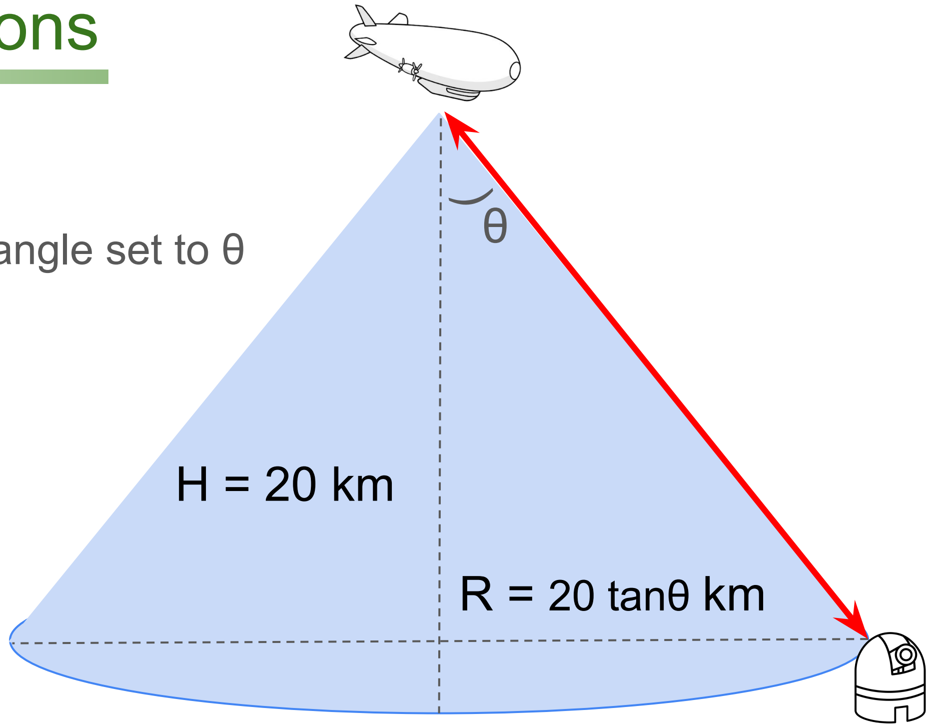




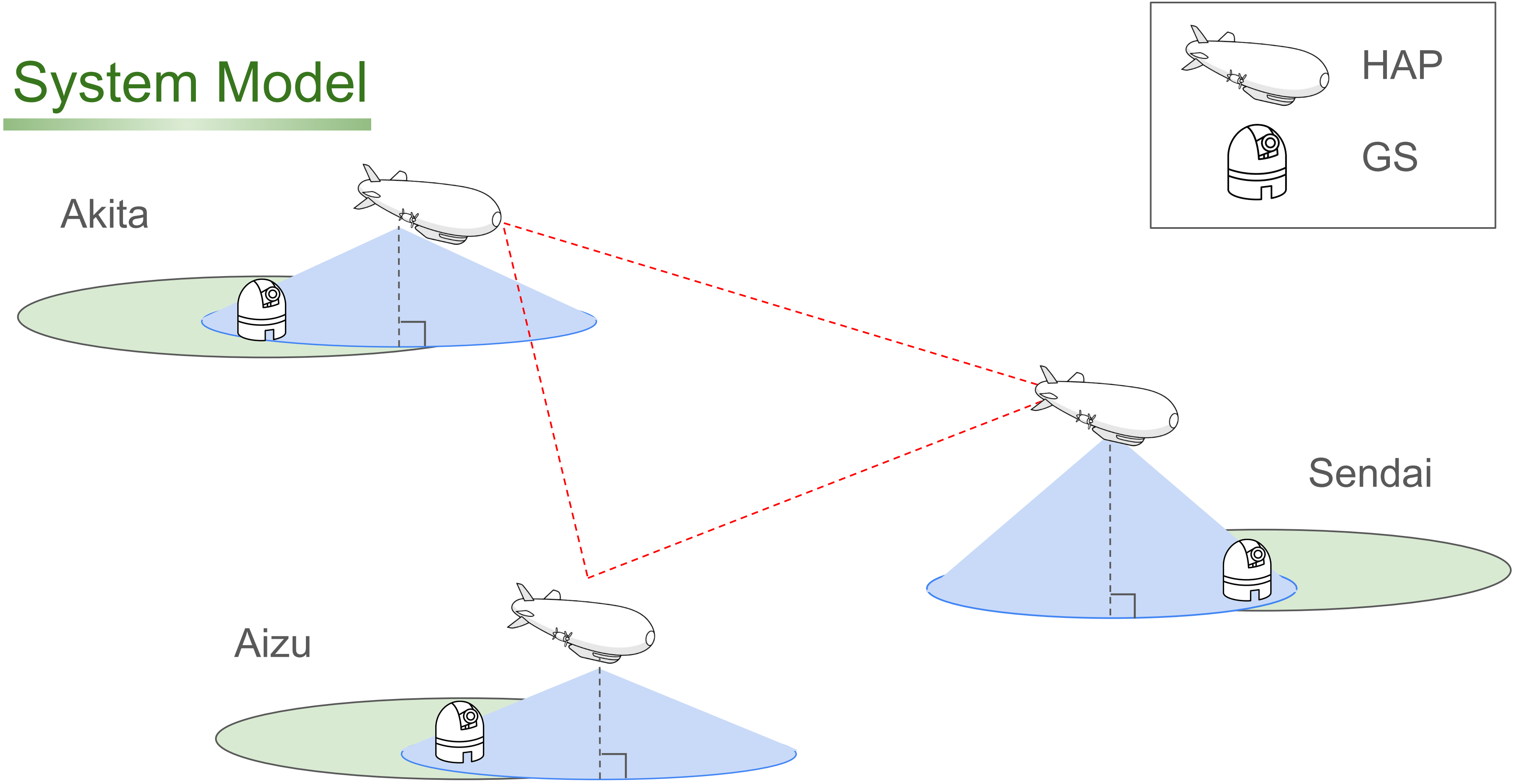
Defining HAP candidate positions

When Zenith angle is θ

- HAP altitude H = 20 km; maximum zenith angle set to θ
- Convert coverage to horizontal distance:
  $R = H \cdot \tan\theta$



System Model



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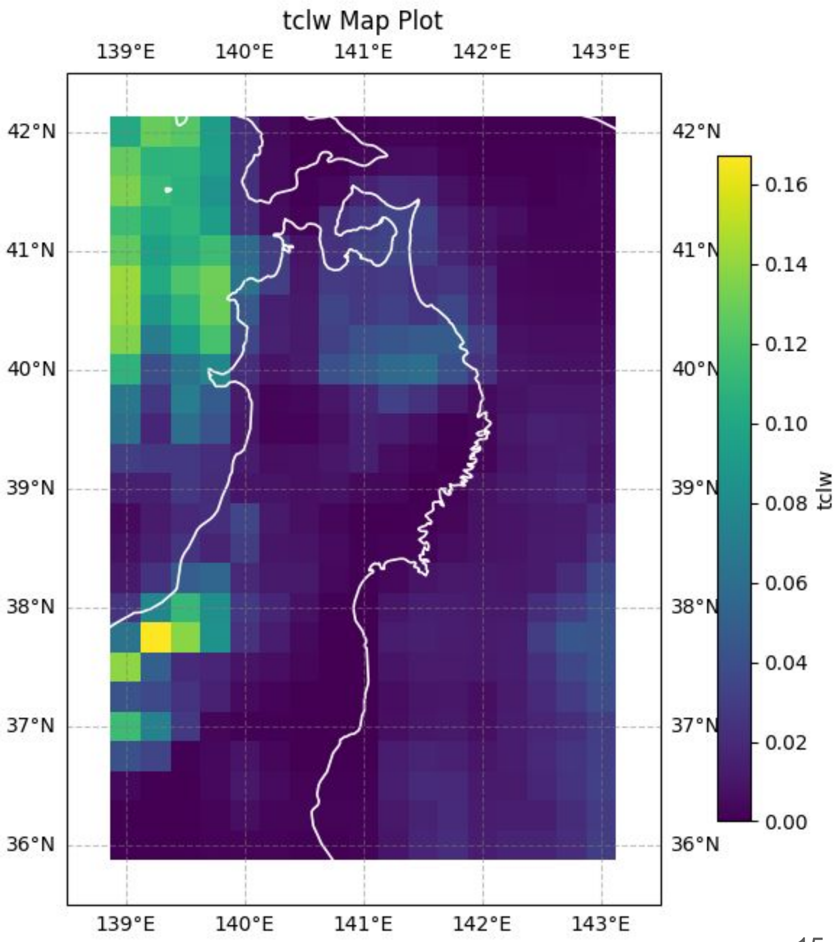
Pixel Data

Period	1 year (2024)
Variable	Cloud (CLWC)
Region	Tohoku, Japan



Problem Formulation

$$\text{P1:} \quad \min_{(x_i, y_i)} \quad w_1 \cdot \mathrm{CLWC}(x_i, y_i) + w_2 \cdot d(x_i, y_i)$$
$$\text{s.t.} \quad (x_i, y_i) \leq (x_{\max}, y_{\max})$$
$$\text{where } w_1, w_2 \text{ are weights: } \quad w_1 > w_2 \text{ (CLWC priority)}, \quad w_1 < w_2 \text{ (distance priority)}$$

- Minimize the weighted sum of cloud (CLWC) and distance (d): $w_1 \cdot \mathrm{CLWC} + w_2 \cdot d$
- Decision variables: HAP positions (x_i, y_i)
- Constraint: positions stay within the bound $(x_i, y_i) \leq (x_{max}, y_{max})$
- Weights set the priority: $w_1 > w_2$ favors low cloud, $w_1 < w_2$ favors short distance

Algorithm

Input: GS positions (6 cities), monthly-mean cloud, coverage radius R
Output: HAP positions (x_i, y_i) minimizing $w_1 \cdot \mathrm{CLWC} + w_2 \cdot d$

Step1: Build candidate sets
For each city, collect grid cells within radius R of its GS.
→ a candidate set per city

Step2: Enumerate combinations
Pick one candidate cell per city
→ repeat the steps below for all n

Step3: Score each combination
Cloud: sum of CLWC over the chosen cells
Distance: open-path length (open TSP)

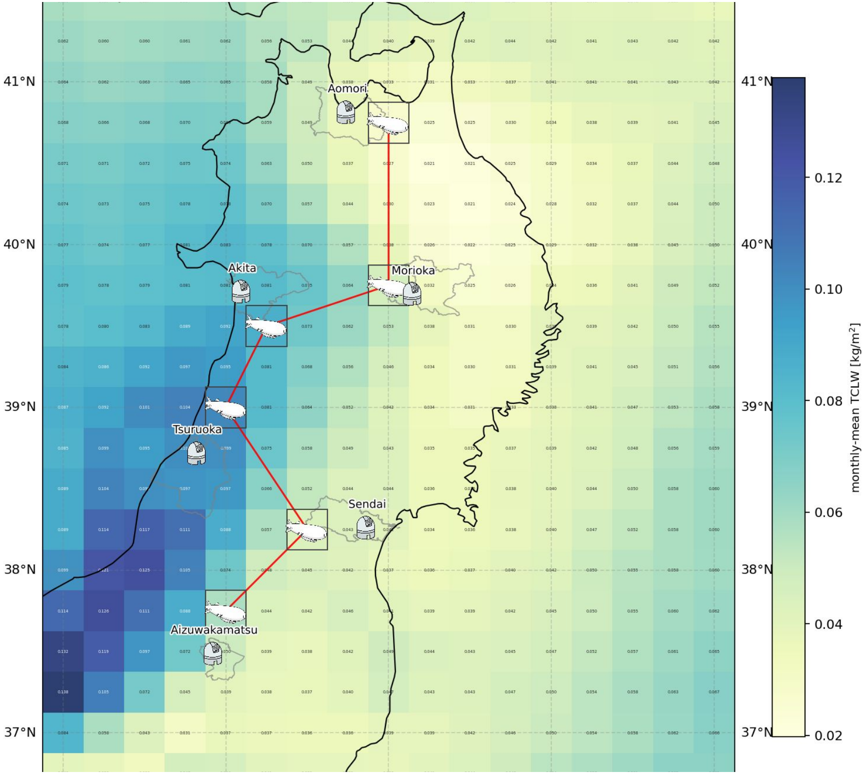
Step4: Normalize & combine
Scale CLWC and d each to [0,1], then
$\mathrm{objective} = w_1 \cdot \mathrm{CLWC} + w_2 \cdot d$

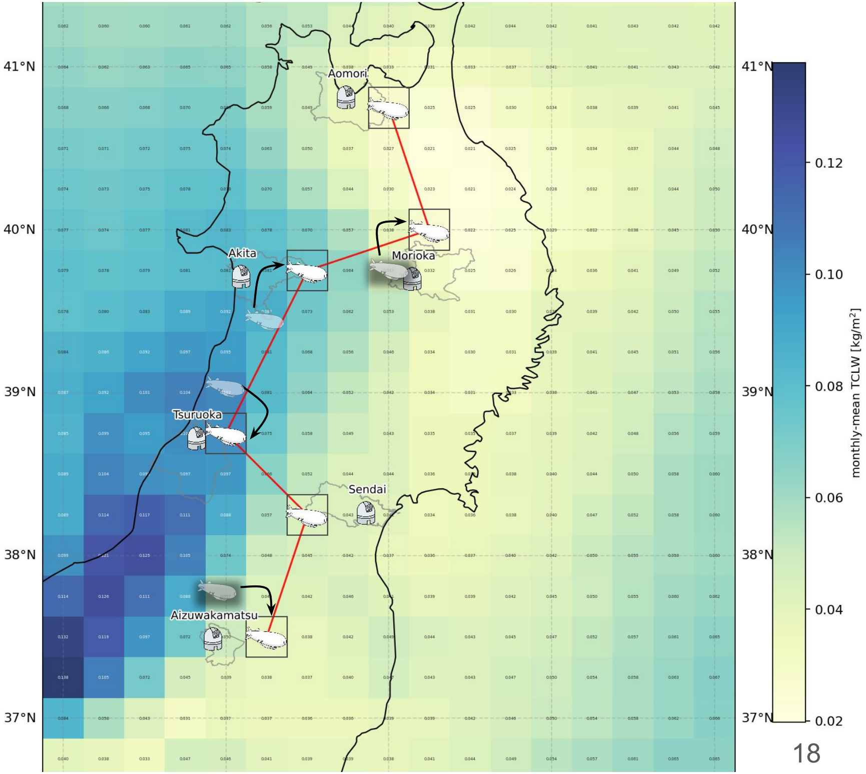
repeat Step 3-4 for all n combinations

Step5: Select
Output the combination with the minimum objective
(guaranteed optimal — exhaustive)

Result

Cloud-aware placement shifts 4 HAPs (Akita, Morioka, Tsuruoka, Aizu) toward clearer cells





Consideration

	Only Distance (Left figure)	Include cloud (Right figure)
CLWC (kg/m^2)	0.3641	0.3103 (-15%)
Distance (km)	405.6	431.9 (+26 km)

Interpretation

- Cloud and distance are trade-off

Future work

1. Convert CLWC to the actual secret key rate (SKR): evaluating how the 15% CLWC reduction affects into SKR.
2. Extend the simulation from monthly to yearly, capturing seasonal variations in cloud cover.
3. Simulate group key generation across the 6 cities.

Thank you for your attention !